

CSC-220: Neural Networks and Deep Learning

Lecture 00 - Introduction to Deep Learning¹

September 1st, 2026

¹Image content: https\protect\protect\protect\leavevmode@ifvmode\kern+.2222em\relax/\deeplearningwithpython.io/chapters/chapter01\protect_what-is-deep-learning/

What this course is about

Neural Networks and Deep Learning provides a structured, hands-on introduction to modern deep learning using Python.

The course connects foundational ideas to major application areas:

- neural networks, tensors, optimization, and backpropagation;
- computer vision and convolutional neural networks;
- language models, attention, and Transformers;
- generative models, including VAEs and diffusion models;
- privacy, best practices, and high-performance computing.

The course progresses from foundations to modern models

The learning sequence is cumulative:

- build the mathematical and computational foundations;
- apply neural networks to images and text;
- understand how modern generative architectures work;
- examine robustness, privacy, and responsible deployment;
- critically analyze current deep-learning research.

By the end of the course, students should be prepared for further study or work in academic and industry settings involving deep learning.

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The most recent Hofstra student to work with me, throughout last semester and the summer, is now preparing to submit a 16-page paper to a leading IEEE journal and is in active discussions with NVIDIA about a research engineer position.

Prerequisites and course logistics

Expected background

Calculus, including multivariable calculus

Familiarity with linear algebra

Basic probability and programming experience are desirable

Instructor and support

Marco Romanelli — marco.romanellihofstra.edu

Office hours: Tuesdays, 1:15–2:30 PM, SIC 215

Textbook purchase is not required; suggested references accompany the schedule

Participation and project presentations are central

Attendance and participation

Attendance is recorded at the beginning of class, although it is not mandatory.

Active participation is expected and contributes to the assignment/participation grade.

Assignments must be submitted on time; late submissions are not accepted (let me know ahead of time if most of you need an extension).

Project presentations

Two tentative presentation dates appear in the course schedule.

The project requires a comprehensive study of a recent, relevant deep-learning research paper. This can be conducted individually or in pairs, with the instructor's approval. Please, communicate whether you will work alone or in pairs by the end of the second week of classes.

The course grade has two components

Component	Weight
Assignments + Participation	60
Research-Paper Project	40

The project is evaluated through:

- depth and accuracy of the analysis;
- understanding of the paper's contributions and methods;
- clarity of the in-class presentation;
- ability to discuss implications and engage critically with the work.

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This means that, given n assignments, with scores between 0 and 100, the resulting grade will be calculated as follows:

$$\text{Final Grade} = 0.6 \cdot \frac{\sum_{i=1}^n \text{Assignment}_i}{n} + 0.4 \cdot \text{Project Score}$$

A grade of C- or better is required to pass

Grade	Percentage Range
A	94–100
A-	90–93
B+	87–89
B	84–86
B-	80–83
C+	77–79
C	74–76
C- (Passing)	70–73
D+	67–69
D	64–66
D-	61–63
F	0–60

Professional conduct protects the learning environment

Respectful behavior and professionalism are expected at all times.

Students are expected to take notes and engage during class.

Unauthorized recording, photography, or disruptive electronic-device use violates the classroom conduct policy and may result in disciplinary action.

Plagiarism and other forms of academic dishonesty are serious offenses.

Students must follow Faculty Policy Series #11 or #11G, as applicable.

Course materials support both theory and practice

Shared course resources include:

- lecture notes in slide format;
- coding problems and hands-on exercises;
- autograded coding assignments;
- references for additional theoretical and practical depth.

These materials are sufficient for strong performance in the course. Students seeking more depth may consult the suggested textbooks, library materials, and online resources.

Office hours work best with a focused question

Office hours are held Tuesdays from 1:15 to 2:30 PM in SIC 215.

Meetings are limited to 15 minutes per student.

Appointments are first-come, first-served; sign up on the office door.

Walk-ins may be accommodated, but scheduled students receive priority.

Before attending, complete the form in “Resources for Students” and identify the specific concepts or materials you want to discuss.

If you cannot attend, provide at least 24 hours’ notice so another student can use the time.

Schedule: September

Date	Topic	Reference	Notes
Tue, Sep 1, 2026	L00 — First Day: Course introduction; Syllabus overview; Introduction to deep learning	Chollet, §§1.1–1.8	GradeScope Init
Tue, Sep 8, 2026	L01 — Mathematical Building Blocks: Tensors and operations; MLPs and backpropagation	Chollet, §§2.1–2.4	Assignments #1.1 and #1.2 Release
Tue, Sep 15, 2026	L02 — Probability and Measures of Information	Hastie et al., §6.6.3; Polyanskiy & Wu, §§1–3; Cover & Thomas, §§2.1–2.10	Assignment #2 Release
Tue, Sep 22, 2026	L03 — Convolutional Neural Networks	Venkatesan & Li, §§2.1–2.10	Assignment #3 Release
Tue, Sep 29, 2026	L04 — Variational Autoencoders: From autoencoders to the reparameterization trick	Kingma & Welling	Assignment #4 Release

Schedule: October

Date	Topic	Reference	Notes
Tue, Oct 6, 2026			Mid-Semester Break
Tue, Oct 13, 2026			Presentation #1
Tue, Oct 20, 2026	L05 — Adversarial Attacks on Image Classification	Li et al., §§3–4	Mid-Semester Advisories; Assignment #5 Release
Tue, Oct 27, 2026	L06 — Text Classification	Tunstall et al., §2	Assignment #6 Release

Schedule: November–December

Date	Topic	Reference	Notes
Tue, Nov 3, 2026	L07 — Language Models and the Transformer: Language modeling; Transformer architecture; Attention	Tunstall et al., §3	
Tue, Nov 10, 2026	L08 — Generative Deep Learning: Text generation; GPT and beyond	Tunstall et al., §4	Assignment #8 Release
Tue, Nov 17, 2026	L09 — Generative Deep Learning: From VAEs to diffusion models	Lai et al., §§1–5	Assignment #9 Release
Tue, Nov 24, 2026			Presentation #2
Tue, Dec 1, 2026	L10 — Privacy in Deep Learning and Data Science	Dwork & Roth, §§1–4	Assignment #10 Release
Tue, Dec 8, 2026	L11 — Best Practices and High-Performance Computing		

What to do next

Confirm access to the course website and GradeScope.

Review the syllabus and tentative schedule.

Note the tentative presentation dates: October 13 and November 24.

Prepare to work regularly in Python and complete the coding assignments on time.

Begin thinking about recent deep-learning papers that may be suitable for the project.

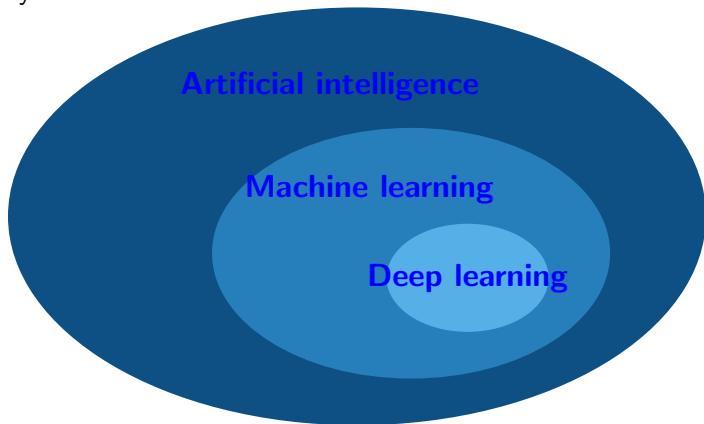
The course begins by connecting tensors, multilayer perceptrons, and backpropagation, the core building blocks used throughout modern deep learning.

What is Deep Learning?

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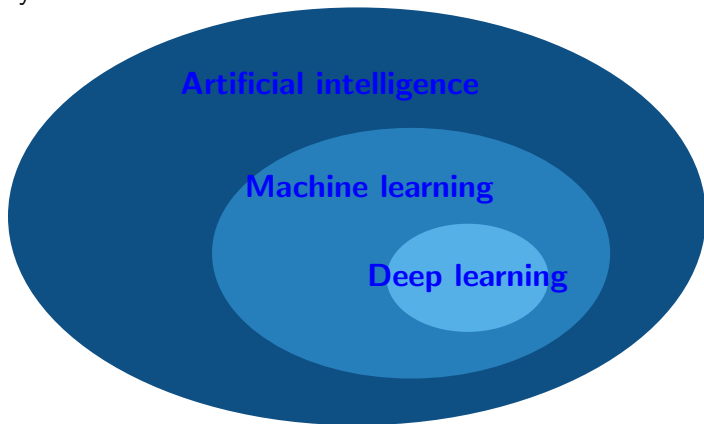
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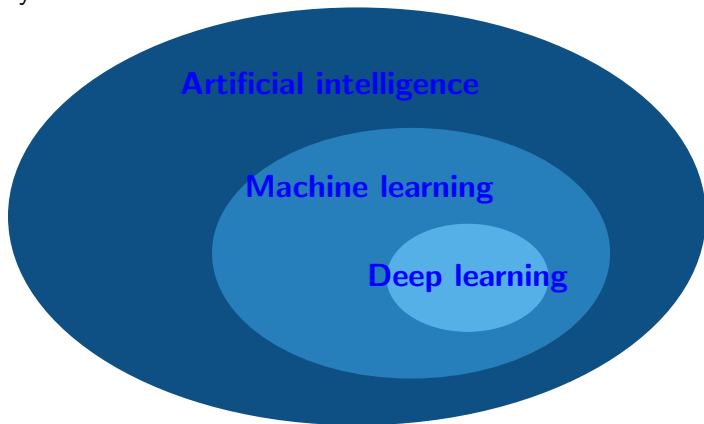
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- Uses neural networks with many layers
- Learns hierarchical representations
- Particularly effective for large datasets

What is Machine Learning?

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Before we can understand deep learning, we need to understand machine learning.

From Programming to Machine Learning

Traditional approach:

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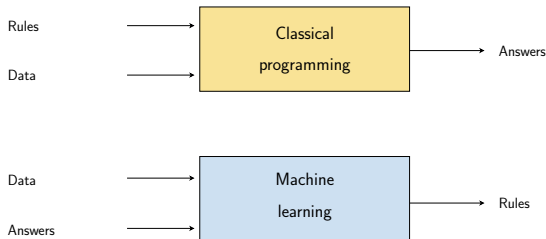
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- e.g., “landscape”, “food”

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- Relies on advances in software and hardware

What Do Machine Learning Algorithms Need?

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Three essential components:

Three Components of Machine Learning

Input data points

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Speech recognition → audio signals

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Image tasks → pictures

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Examples of expected outputs

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Image tasks → pictures

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Images → labels (“dog”, “cat”, etc.)

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- This adjustment process is called *learning*

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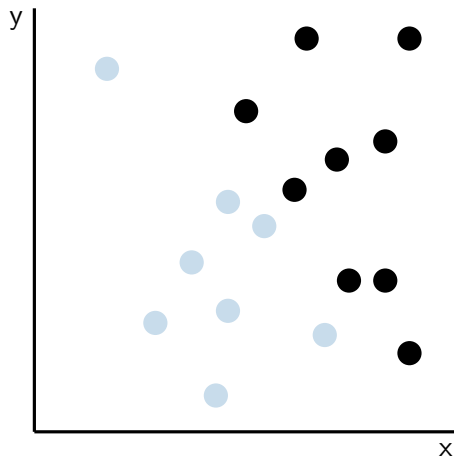
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Machine learning finds representations suited to the task

- Transforms data into more useful forms

Example: Classifying Points in 2D



A Simple Classification Task

We are given points in (x, y) space

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Each point is labeled: black or blue

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Learn a function that maps $(x, y) \rightarrow \text{color}$

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Performance measure

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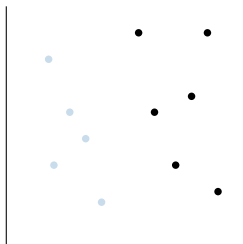
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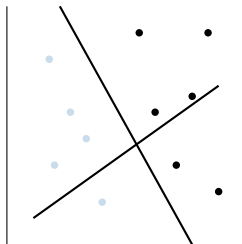
Percentage of correctly classified points

From Raw Data to Better Representation

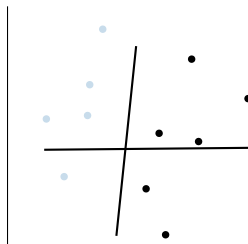
1: Raw data



2: Coordinate change



3: Better representation



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In the new coordinate system, each point has new coordinates

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Next: what makes *deep learning* special?

What Does “Deep” Mean?

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Focus:

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Learning *representations* from data

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Through successive layers

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Each layer builds more meaningful representations

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- Hierarchical representation learning

Deep vs Shallow Learning

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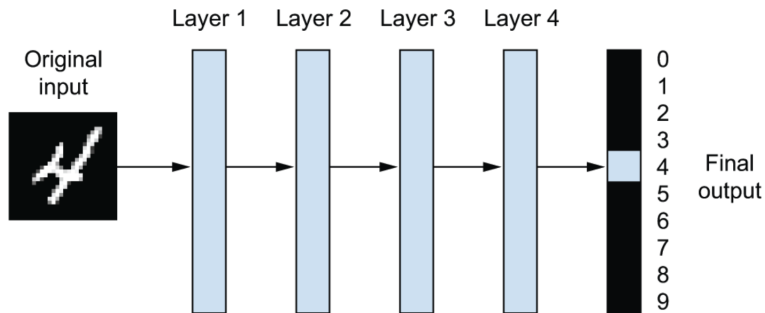
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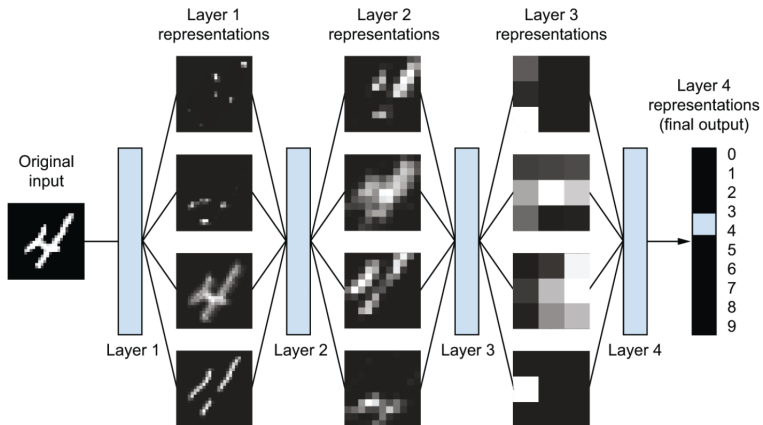
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These are often called *shallow learning*

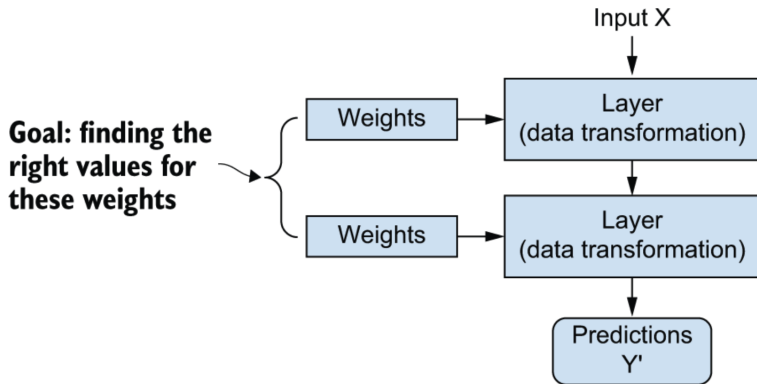
Learning Patterns



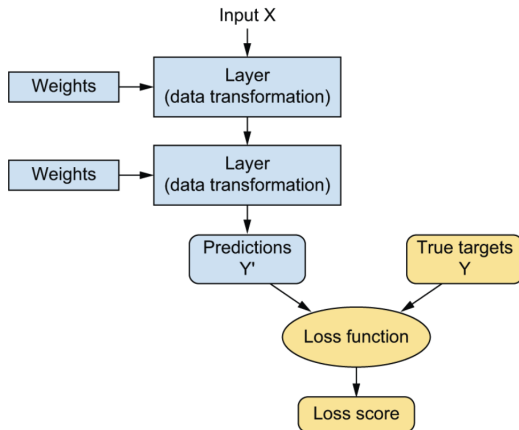
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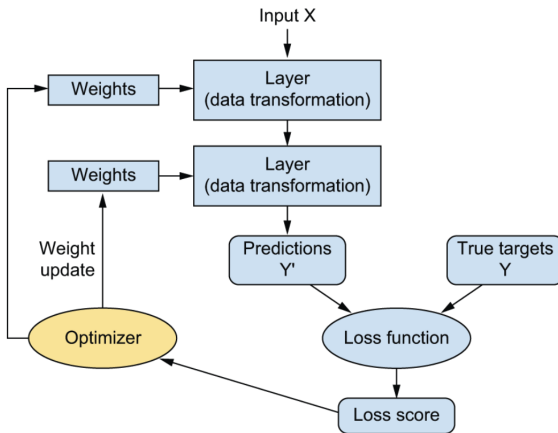
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Scalability — Deep learning scales efficiently with data and compute, leveraging GPUs and parallel processing to handle very large datasets.

Simplicity — It reduces the need for manual feature engineering by automatically learning useful representations, enabling simpler end-to-end models.

The Age of Generative AI

Generative AI systems (e.g., ChatGPT, Gemini, Claude, Midjourney) can produce text and images from simple prompts, blurring the line between human and machine creativity.

They are powered by large *foundation models* trained to predict or reconstruct data (e.g., next-word prediction, image reconstruction), using vast amounts of unlabeled data (self-supervised learning).

These models scale to enormous sizes (billions of parameters, massive datasets), enabling unprecedented performance.

As a result, they act as flexible knowledge systems that can be applied to many tasks with minimal retraining—often just through prompting.